

RUIKAI YANG

Atlanta, GA · ✉ ryang435@gatech.edu · in linkedin.com/in/ruikai-yang · 🌐 kyree-yang.github.io

EDUCATION

Georgia Institute of Technology

M.S. in Computer Science (MSCS)

Aug 2026 – Dec 2027 (Expected)

Atlanta, GA

University of Michigan, Ann Arbor

B.S.E. in Computer Science (Dual Degree), Summa Cum Laude | GPA: 4.00/4.00

Aug 2024 – May 2026

Ann Arbor, MI

Shanghai Jiao Tong University

B.Eng. in Mechanical Engineering | GPA: 3.85/4.00

Sep 2022 – Aug 2026

Shanghai, China

EXPERIENCE

TikTok (ByteDance)

Software Engineer Intern, Intelligent Creation

May 2026 – Aug 2026

San Jose, CA

Autonomous Bug-Fixing Platform | LLM Agents, Full-Stack (React 19 / NestJS / PostgreSQL), Swift/Kotlin, TypeScript

- Built a platform where headless LLM agents root-cause a production iOS/Android defect, patch it, open a real merge request and drive CI green, pausing only for on-device verification — lifting audited first-attempt fix rate **17%** → **70%**, 63 of 88 defects closed end-to-end.
- Rolled out to **80+ QA engineers** who now close defects self-serve with no developer pulled in, cutting mean turnaround per defect from ~2 engineer-days to ~2 hours.
- Built the distributed execution plane — a daemon over 7 DB-backed runners and 10 concurrent agents — with compare-and-swap job claiming, per-ticket **git worktree** isolation, and an SSE pull channel cutting click-to-action latency **90s** → **1–2s**.
- Hardened autonomy with a 20-step state machine enforced out-of-band by 7 hooks gating every transition on artifact existence, orphan-process adoption (**47%** of runs lost their driver), and a SWE-bench-style repro benchmark from reverted real fixes.
- Traced a long-standing job-interleaving corruption to a concurrency gate that had **never once fired** (its process count returns 0 on BSD); replaced it with atomic CAS claiming and UUID-keyed process markers, verified on a 9-scenario race harness.

Design Lab — Agentic Design-to-Code Workflow | LLM Agents, DAG Orchestration, Context Engineering, iOS/Android

- Split each design-to-code task into a **seven-stage DAG**: heavy stages run in isolated subagents that keep ~950 lines of design rules out of the main context, lifecycle hooks auto-register every node so no step is silently skipped, and each node ships an on-stop gate blocking the agent until its artifacts exist.
- Built the localization and weak-network nodes on that spine: one shared **328-string** table across 5 locales replacing iOS JSON + dual Android XML, delta-only machine translation cutting a 5-minute full resync to seconds, and a review subagent auditing RTL mirroring, locale formatting and text expansion behind a deterministic checker.

RESEARCH EXPERIENCE

Robotics and Optimization for Analysis of Human Motion (ROAHM) Lab

Research Assistant | Advisor: Prof. Ram Vasudevan

Sep 2024 – Apr 2026

University of Michigan

- Engineered optimized C++ modules for analytical Jacobian computation inside a PDE-based trajectory optimization framework, cutting solver execution time by **40%** on average (Co-author, *IROS 2026*).

SELECTED PROJECTS

AgentOS — Domain-Specific Agent Workflow Planner | Systems Lead · LangChain, Python, YAML DSL

May – Aug 2026

- Led the planning layer of an industrial agent platform (capstone, sponsor Hangzhou AIMS) turning a plain-language prompt into an executable **YAML workflow DAG**: a meta-agent plans against a retrieval-scoped skill tree loading one branch instead of the whole library, calls knowledge-base tools instead of hallucinating facts, and asks the operator only at plan time.
- Froze each plan behind three validity gates so execution is deterministic and drift-free, after validating the paradigm by reproducing LLMCompiler end-to-end — cutting wall-clock **15.3s** → **9.3s** (1.64× speedup) against a serial ReAct baseline at equal-or-better accuracy — and built the WorFBench evaluation arm scoring generated workflows by node- and graph-level F1 against hand-authored gold DAGs.

Distributed Internet Search Engine | C++, Distributed Systems, MapReduce, Linux

- Architected a distributed search engine on a **12-node** Linux cluster (GCP VMs + local) indexing 30M+ pages, with multi-threaded crawlers and crash recovery sustaining high-throughput crawling over 240GB of HTML.
- Designed a hash-partitioned inverted index sharded across servers with stop-word filtering and delta/variable-length encoding (UTF-8 cut on-disk footprint ~50%), plus a networked query service whose central aggregator merges and re-ranks partial results — profiling cut tail query latency from ~8s to under **2s** under concurrent load.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, TypeScript/JavaScript, SQL, Bash

AI & Agents: LangChain/LangGraph, Claude Agent SDK, MCP, tool & skill design, context engineering, multi-agent DAG workflows, sandboxing & process isolation, evaluation & benchmarking, PyTorch

Systems & Tools: Distributed Systems, Concurrency, Docker, PostgreSQL, React, NestJS/Node.js, REST/SSE, Git, Linux

HONORS

MCM Outstanding Winner (Top <1%, 2024) · China National Scholarship · James B. Angell Scholar & University Honors (UMich)